POLITECNICO MILANO 1863 | SCHOOL OF INDUSTRIAL AND INFORMATION ENGINEERING

Computational Fluid Dynamics – A.A. 2024-2025

SR-71 "Blackbird" Inlet Flow Analysis across Different Mach Regimes

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1. Basis to the study of the supersonic inlet of the SR-71

1.1. Scientific Context and Background

The Lockheed SR-71 "Blackbird" remains one of the most sophisticated supersonic aircraft ever developed, capable of sustained flight at Mach 3.2. The aircraft propulsion system relied on two Pratt & Whitney J-58 engines, which transitioned into quasi-ramjet mode at higher speeds. A critical component enabling this performance is the inlet, which operated as a **mixed external/internal compression system**. The inlet geometry features a translating conical spike, whose position was adjusted progressively to control the location of the terminal shock and optimize airflow into the engine. The inlet system was designed to:

- Maintain **self-starting capability** across a range of speeds up to Mach 3.2, by satisfying the Kantrowitz limit for subsonic flow at the throat.
- Manage a series of **reflected shock waves** within the duct to achieve the necessary pressure and temperature ratios for efficient engine operation.
- Prevent flow separation and *unstart* through the use of **spike translation** and bleed systems.

The spike retraction, which began at Mach 1.6 and continued until Mach 3.2 at a pace of 4.1 cm per 0.1 Mach, allowed the inlet to adapt to the increasing free-stream Mach number. This process increased the captured stream-tube area and positioned the shock system accordingly, in order to slow down the flow directed to the engine.

An additional feature of the inlet system, as shown in Figure 1, was a 25.4 cm wide porous surface, located around the spike along its point of maximum curvature, which served as a bleeding system [7]. The main purpose of the spike bleed was to prevent flow separation by removing low-momentum fluid near the wall, therefore enhancing flow stability in case of unsteady perturbations.

The last component relevant to this study is a device referred to as 'shock trap', which was located on the inner cowl. Its purpose was to absorb another portion of mass flow and induce a normal shock wave, ensuring a subsonic flow in the diffuser before the engine face. Overall, these active control devices played a crucial role in regulating the flow-field of the SR-71 inlet.

An issue with this design was that for some flight conditions it would occur that both the convex geometry of the spike and the adverse pressure gradient after each shock reflection forced the turbulent boundary layer to enlarge, consequently increasing the displacement thickness and the low-momentum fluid region, increasing dissipation. For mixed compression intakes, if the upstream, external mass flow is greater than the downstream, internal one, the intake will unstart as a result of this supersonic chocking. The characteristic normal shock of the supersonic inlet will then propagate upstream, exiting the duct, until the mass flow mismatch reaches equilibrium, as discussed by Anderson [1].

This phenomenon led to an enormous asymmetric increase in drag and a drop in thrust, which could cause loss of control of the aircraft.
To prevent these unstart conditions, the main control action engineered for the SR-71 engine was the adjustable spike position.

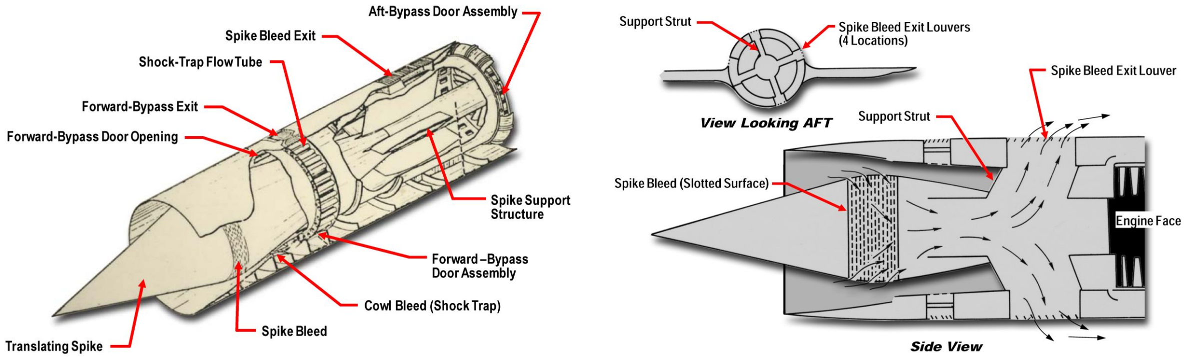


Figure 1: Illustrative sketches showing the geometry of the internal components of the inlet.

1.2. Research Questions

The objective of this work is to analyse the flow of the inlet mounted on the Lockheed SR-71 "Blackbird" at selected supersonic Mach numbers using Computational Fluid Dynamics (CFD) simulations. More in detail, the analysis focuses on understanding the flow behaviour in the mixed compression inlet for different spike positions, corresponding to Mach numbers 1.7, 2.0, 2.5, 3.0 and 3.2.
Similar to the SR-71 studies conducted by J.R. Coleville [4], the simulations in this work employ the Euler equations to model inviscid, compressible flow, providing a suitable starting point due to their low computational cost.
To validate the accuracy of the inviscid simulations, a viscous comparison is necessary: this is addressed through Reynolds-Averaged Navier-Stokes (RANS) simulations, which reveal the effects of the boundary layer on the internal flow, particularly boundary layer thickening, shock-boundary layer interactions (SBLI) and the resulting losses in mass flow and total pressure.
Unlike Coleville's work, which investigates an extended operational regime, this study focuses on the steady-state response of the inlet at specific Mach numbers that reflect the actual flight conditions of the aircraft.
The results aim to provide insights into the self-starting characteristics and overall flow behaviour of the inlet, contributing to a broader understanding of mixed compression inlet operation in supersonic regimes.
To achieve this set of goals, the following research questions were posed:

- *What is the behaviour of the SR-71 inlet across different Mach number regimes? What is the internal flow structure and shock positioning?*
- *Do the numerical results match the findings of independent research or experimental results?*
- *How does the flow change when accounting for viscous effects in a RANS simulation? In particular, how much do the pressure recovery and mass flow differ?*
- *Are the results of the RANS simulations consistent with the expected physics of the problem? What potential issues, such as flow unstart, might arise?*
- *Is the use of inviscid Euler equations justified for the investigation of this kind of problem?*

In order to simplify computational requirements and ensure manageable simulation runtimes, 2D axisymmetric simulations are employed, leveraging the SR-71 inlet annular design.
All simulations are conducted using the SU2 open-source CFD software [6] on personal computing resources.

1.3. Underlying Hypothesis

This study is based on the assumption that the SR-71 inlet flow can be accurately modelled using certain simplifying hypotheses. Specifically:

1. The flow field is considered to be in **steady state**.
2. The air is modelled as an **ideal and polytropic gas**, with no dissociations.
3. The **angle of attack** and **side-slip** of the incoming flow are null.
4. While employing **Euler equations**, their defining hypotheses are assumed to be valid (inviscid flow, zero heat flux).
5. The solid walls can be modelled as **rigid and adiabatic**.
6. The flow can be considered **axialsymmetric** with respect to the spike axis.
7. The simulation of the **spike bleed is neglected**, due to the lack of information that is needed to set the correct boundary conditions for it.

1.4. Significance of the Work

This work serves as a benchmark for high-speed inlet CFD analysis using limited computational resources. Employing an axisymmetric model, it highlights the critical aerodynamic features of the SR-71 inlet while balancing computational cost and accuracy. The results are particularly relevant for future supersonic and hypersonic inlet designs, where understanding shock wave behaviour and self-starting limits remain critical. Additionally, this study demonstrates the application of *SU2* for solving complex aerospace problems with open-source tools.

2. Numerical Simulation Methodology

2.1. Problem Geometry

The exact geometry of the inlet couldn't be recovered: due to the military background of the aircraft, no public blueprints are easily available. Coleville [4] managed to gather enough information from trustworthy sources and infer others by reverse-engineering. The results of his research are summed up in Figure 2 and are also the reference for this study.

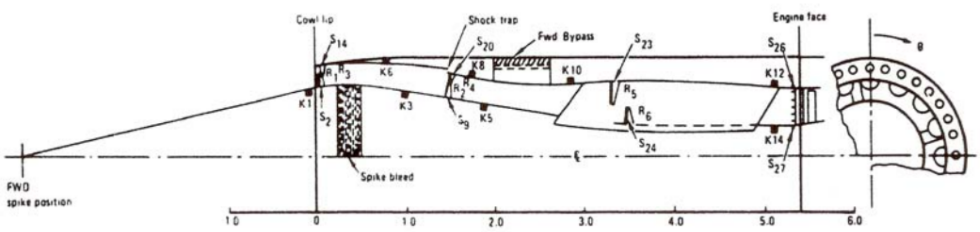


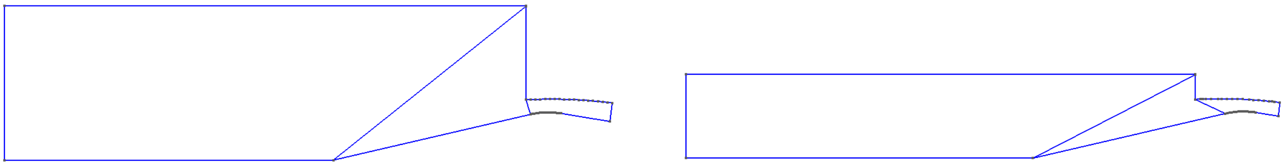
Figure 2: Sketch used to derive the geometry.

The unit length of the horizontal axis in Figure 2 is one cowl radius, which was measured at 0.74 m from other sources. The image was imported into `Autodesk Fusion`™ (a CAD software optimized for reverse engineering applications): there it was rescaled accordingly to the correct proportions and exported as an `.stl` file. A custom `MATLAB`™ script was used to extract the coordinate points which were then used to create the `.geo` file.

Due to the fact that the spike of the SR-71 inlet can be moved inwards or outwards depending on the specific flight condition, a different domain geometry was used at each investigated Mach number in order to always study the system at its design position.

However, because of the impossibility of correctly modelling the shock trap, the problem only investigates the domain up to that point, also excluding the subsonic divergent diffuser all the way to the engine face.

Finally, the height of the different domains was varied according to each Mach number to avoid reflection of the first shockwave on the upper domain boundary due to numerical effects. Two examples are shown in Figure 3.



M_{∞}=1.7 M_{∞}=2.5

Figure 3: Domain geometry for two different flight conditions.

2.2. Mesh Construction

The mesh construction process was tailored for both Euler and RANS simulations. In order to efficiently resolve critical flow features, a hybrid approach was adopted, consisting of structured and unstructured meshes. The mesh generator software of choice was **gmsh** [8].

Euler Mesh For the inviscid simulations, a **fully unstructured mesh** was generated and refined in key areas to capture the shock structures:

- **Inlet external duct region:** Local refinement to accurately resolve the primary oblique shock. The refinement was done around the angled line following the expected inviscid shock angle, derived from oblique shock relations in cylindrical coordinates [5].
- **Inlet internal duct region:** Mesh refinement inside the cowl to capture reflected shocks and steep gradients. Since the exact locations of the shocks and fans were complex and changing for each iteration of the different simulations, a dense and uniform refinement was ensured in this region.
- **Farfield:** A coarser grid in the far-field uniform flow region (to the left of the primary shock) ensured computational efficiency without compromising accuracy.

Figure 4 focuses on a detailed view of the front part of the cowl, highlighting the varying levels of refinement within the mesh while Figure 5 provides an overview of a typical Euler mesh.

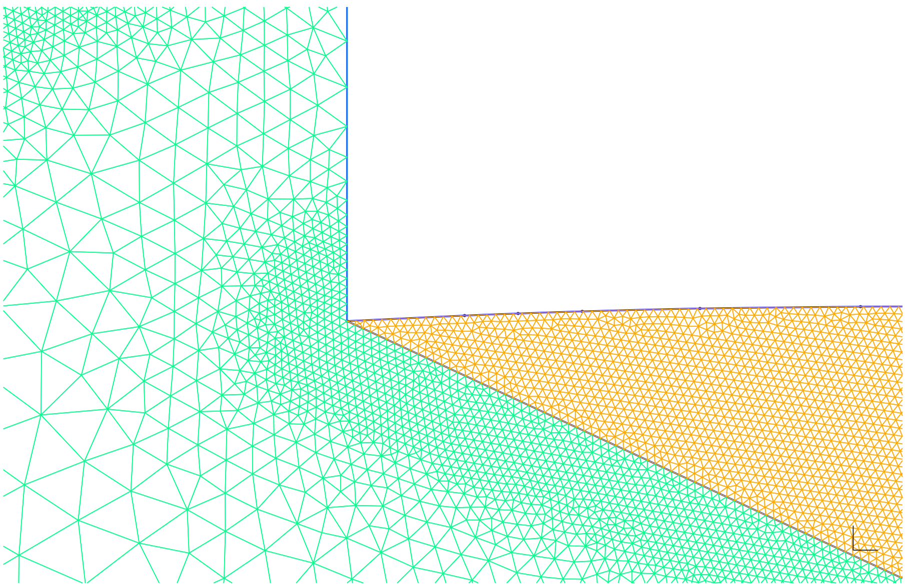


Figure 4: Refinement detail. The mesh was made coarser for image clarity.

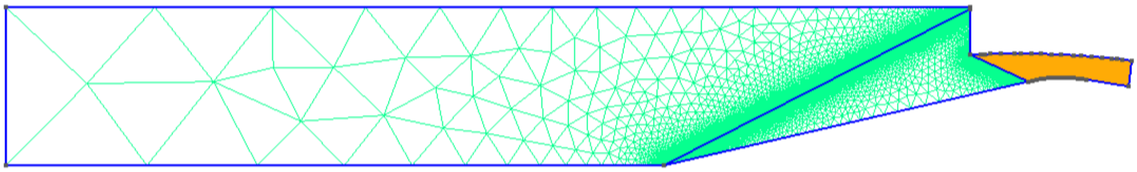


Figure 5: Example of the coarse Euler mesh at $M_\infty = 2.5$.

RANS Mesh For the viscous simulations, a **structured boundary layer mesh** was applied along the solid surfaces (spike, cowl). Initially, the boundary layer cell size in the direction normal to the surface was estimated to correspond to one viscous unit for a flat surface in a stream with the same Reynolds number (which was calculated with an online tool [2]). After the first simulations, this choice was deemed satisfactory. The internal duct volume employed a **structured mesh** consistent with the underlying boundary layer. The remaining domain used an **unstructured mesh**, with targeted refinement near shockwave and wall regions, similar to the Euler mesh. A focus on the interface between the two parts of the domain is visible in Figure 6.

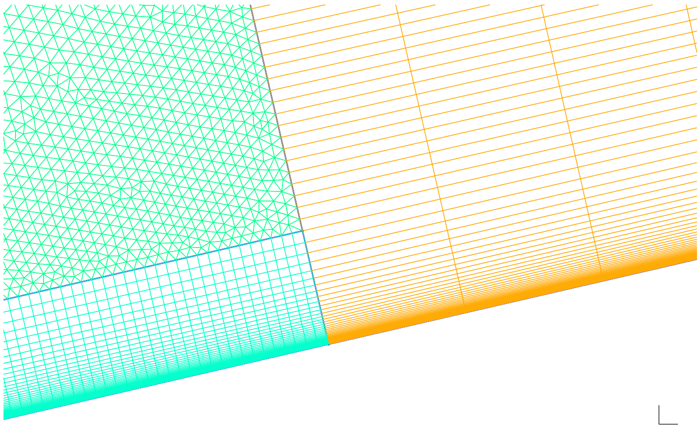


Figure 6: Boundary layer detail at the interface between internal *fully structured* and external *unstructured* mesh. The mesh was made coarser for image clarity.

Grid Levels and Refinement Four grid levels were employed. They are represented together with the corresponding approximated number of elements in Table 1.

	n° Elements (Approx.)	h_f
Coarse	$2 \cdot 10^5$	$2.7 \cdot 10^{-3}$
Medium	$4 \cdot 10^5$	$1.8 \cdot 10^{-3}$
Fine	$9 \cdot 10^5$	$1.2 \cdot 10^{-3}$
Superfine	$1.5 \cdot 10^6$	$0.8 \cdot 10^{-3}$

Table 1: Mesh refinement parameters.

Each grid level was characterized by the smallest cell size allowed for the refinement regions, h_f, while the maximum cell size was left unchanged to $h = 3$. Since the geometry (and thus the domain) changes as the spike is extracted, and the angle of the first shock wave also changes, the exact number of elements varied between simulations at different Mach numbers for the same grid level.

2.3. Boundary Conditions

A fundamental aspect of the study was the accurate representation of domain features through appropriate boundary conditions. For clarity, Figure 7 illustrates how the geometry's edges were labeled.

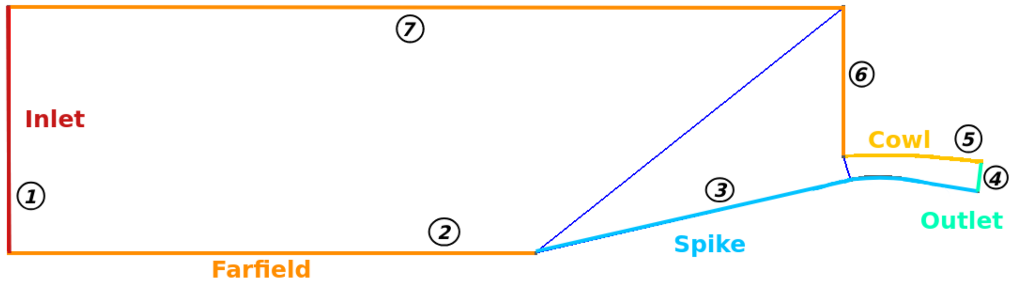


Figure 7

The imposed boundary conditions were:
- **Zero heat flux** on solid surfaces (spike 3 and cowl 5) that translates in:
  - Euler: **non penetration**.
  - RANS: **no-slip**.
- **Supersonic inflow** on the inlet 1.
- **Farfield** on all other surfaces (2, 6, 7), including the outlet 4.

Although intuition might suggest otherwise, the farfield boundary condition was applied to the surfaces that act as outflow regions, namely surfaces 4 and 6. The farfield boundary condition in *SU2* was chosen over the "supersonic outlet" and Riemann conditions due to its implementation. This is explained by the fact that "supersonic outlet" requires fixed flow properties to be assigned and prescribed *a priori*, while there was a lack of documentation for the Riemann condition. In principle, the latter could have worked thanks to its ability to adapt to the flow at each iteration, but the farfield boundary condition proved to be an effective and simpler alternative.
Imposing farfield conditions on surface 6 also ensured that the external shock wave was not artificially reflected downward due to numerical effects, as it would have been if it had impinged on surface 7: this prevented the contamination of the solution.
Finally, using the Euler equations for axisymmetric simulations allowed for the application of a farfield boundary condition on the axis of symmetry.

The free flow properties considered for the simulations were chosen in accordance with ISA standard atmospheric conditions as shown in Table 2, corresponding to the typical cruise condition of the SR-71.

Altitude (ft)	Static Pressure (Pa)	Temperature (K)
90,000	2153	223.15

Table 2: Flow Properties.

2.4. Numerical Methods

2.4.1 Euler Simulation

Euler simulations were a crucial first step in understanding the fundamental flowfield characteristics of the SR-71 inlet: by focusing on inviscid phenomena, they provided insights into shock wave positioning and overall behaviour of the flow. The boundary conditions were carefully defined to replicate the physical environment of the SR-71 at each Mach number as best as possible, under the previously defined assumptions.

The simulation was initialized using a coarse mesh solution and restarted on finer grids to accelerate convergence while ensuring a well-resolved flowfield.

Numerical scheme To solve the governing equations, the **Jameson-Schmidt-Turkel (JST)** scheme was chosen for the spatial discretization of convective fluxes, since the problem is steady-state. This central difference method was deemed the most suitable for the study due to its robustness across a range of flow regimes, including transonic and supersonic conditions, making it versatile and reliable for the study.
The JST scheme is also known for its neutrality, avoiding directional bias in the solution. This is particularly important given the complexity of the domain, which includes both external and internal regions with intricate geometry. Its simplicity, combined with its ability to handle complex problems, ensured an efficient and targeted approach to the simulation. To further enhance performance, the scheme incorporates second- and fourth-order artificial dissipation coefficients (set to 0.5 and 0.02, respectively). This combination allows for accurate resolution of strong shock structures while minimizing spurious oscillations, ensuring the stability and fidelity of the results.
Additionally, an adaptive CFL strategy was employed, starting with a CFL number of 10 and progressively increasing up to 100 as the simulation converged. This approach accelerates convergence in smooth flow regions while preserving numerical stability in areas near shocks. The linear system of equations resulting from the discretization was solved using the FGMRES solver with ILU preconditioning. The solver tolerance was set to 10^{-6} and the maximum allowable iterations are 50. This configuration balanced computational cost and solver precision, which is essential for handling the non-linearities of compressible flow fields.
Finally, the implicit Euler method was chosen for time discretization due to its stability, especially in steady-state simulations of compressible flows. This method ensures numerical stability at high CFL numbers, enabling efficient convergence while managing strong gradients. Its ability to handle the stiffness of governing equations makes it particularly suited for problems like the SR-71 inlet, characterized by high Mach numbers and complex flow interactions. Unlike explicit methods, it allows for larger time steps, reducing computational cost without compromising accuracy. Additionally, its compatibility with adaptive CFL strategies enhances efficiency by accelerating convergence in smooth regions while maintaining stability near shocks and critical flow features.
A brief summary is outlined in Table 3.

Parameter	**Selection**	**Reason**
Scheme	JST	Robustness, Adaptability, Efficiency
CFL number	$0.5 \div 100$	Fine-tuned

Table 3: Euler settings.

Trade-offs The use of Euler-based methodology was guided by the need to balance accuracy and computational efficiency:

- **Advantages:**
  - Captures primary aerodynamic phenomena, such as shock wave interactions and flow compression, without the complexity of viscous effects.
  - Provides a computationally efficient baseline for assessing mixed compression inlet flow characteristics under inviscid assumptions.
- **Limitations:**
  - Neglects viscous effects, which are especially relevant for simulating the boundary layer behaviour and shock-boundary layer interactions (SBLI).
  - Simplifies wall boundary conditions, omitting skin friction and heat transfer effects.

Grid convergence Mesh convergence of each set of simulations was achieved after 4 refinements, with each run reaching a value of the residual on the density field below 10^{-12}, ensuring that the flow

features, including shock structures and pressure recovery, were accurately captured.
In order to study grid convergence, it was necessary to choose a different parameter with respect to the one used by the numerical method to compute the residual. The chosen parameter was the drag coefficient, which in this case only amounted to pressure drag.
Grid independence studies, displayed in Figure 8, demonstrate that the finest mesh produces minimal differences compared to coarser grids, of the order of 1%, thus validating the robustness of the numerical setup. For the purpose of this analysis, the relative error on the C_d was defined as:

$$C_d^{rel} = 100 \cdot \left| \frac{C_d}{C_{d,superfine}} - 1 \right|$$

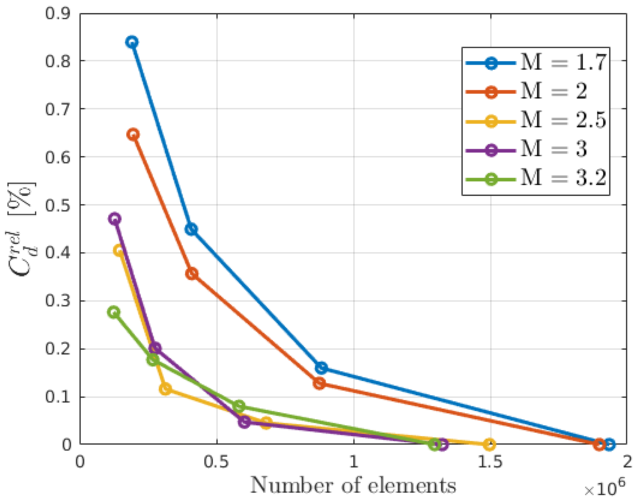


Figure 8: C_d percentage error relative to the finest mesh. Euler cases.

Farfield convergence was not carried out, since the external region of the domain exhibits a uniform flow upstream of the first oblique shock wave. Consequently, any changes occurring downstream of the shock do not influence the solution in the upstream portion of the domain.

2.4.2 RANS Simulation

The viscous simulations were conducted using the Reynolds-Averaged Navier-Stokes (RANS) equations to capture the effects of boundary layers and Shock-Boundary Layer Interactions (SBLI) within the inlet, offering a realistic assessment of the inlet performance.
Due to limited computational resources, the analysis focused exclusively on the case $M_\infty = 3.2$, which corresponds to the design Mach number for the aircraft cruise conditions and is also the one investigated in the RANS simulation conducted by Coleville [4].

Turbulence Model The **Spalart-Allmaras (SA)** model was selected as it is the only turbulence model supported by the adopted current version of SU2 coupled with the condition of axisymmetry. It is however an industry standard, and it proved to be efficient and reliable.
The SA model employs a single transport equation, making it computationally less expensive compared to multi-equation models like the SST. This efficiency is particularly beneficial given the high Reynolds number and the complex geometry of the SR-71 inlet.
The SA model is well-suited for predicting attached and mildly separated flows, as in this case. Its robustness in handling supersonic flows ensured accurate predictions of the boundary layer interactions with shock waves, a critical factor in maintaining inlet stability and preventing unstart. While the model may have limitations in capturing more complex turbulent phenomena (e.g. large-scale separations), its performance in high-speed aerodynamic applications is justified in this study.

Numerical scheme For convective fluxes, **Roe's flux-difference splitting scheme** was employed to ensure robust and accurate resolution of shock waves and shock-boundary layer interactions: since it is an upwind method, it is particularly well-suited for supersonic flows.
This scheme is also useful for viscous flow simulations, offering a balance between computational efficiency and accuracy. Similarly to the JST scheme, Roe's method is highly effective in handling complex domains and a wide range of flow conditions, including transonic and supersonic regimes. Its strength lies in resolving strong gradients, such as oblique shocks and reflected shock systems within the inlet.
Despite its advantages, Roe's scheme can introduce oscillations in regions with steep gradients, as observed in the results. To mitigate this issue, a slope limiter was employed to stabilize the solution and prevent non-physical behaviours, ensuring the reliability of the outcomes. The scheme was augmented by a MUSCL (Monotonic Upwind-Centered Schemes for Conservation Laws) approach to achieve second-order spatial accuracy, preserving sharp shock structures without introducing excessive numerical dissipation.
The **Venkatakrishnan slope limiter** was applied. This limiter is ideal for maintaining solution accuracy in regions with steep gradients, ensuring smooth transitions in the flowfield. Additionally, similarly to the Euler inviscid simulations, the use of implicit Euler time discretization in the RANS simulations addressed the added complexity of viscous effects and turbulence modelling. The stiffness introduced by the coupling of the Reynolds-Averaged Navier-Stokes equations with the Spalart-Allmaras turbulence model required a method capable of maintaining stability while resolving fine-scale boundary layer phenomena and shock-boundary layer interactions (SBLI).
An adaptive CFL strategy was employed to optimize convergence. Starting with an initial CFL value of 0.1, the solver dynamically adjusts the CFL number up to 50, balancing stability with a steep down factor and computational efficiency, allowing optimal handling of SBLI and non-linearities of compressible viscous flows.
A brief summary is outlined in Table 4.

Parameter	**Selection**	**Reason**
Scheme	ROE	Accuracy, Efficient
CFL number	$0.1 \div 50$	Fine-tuned

Table 4: RANS settings.

Trade-offs The RANS-based methodology has several advantages and limitations, which are summarized as follows:

- **Advantages:**
  - **Detailed boundary layer analysis:** Captures viscous effects, SBLI, and flow separation accurately.
  - **Efficiency of the SA model:** provides reliable turbulence modelling with a lower computational cost compared to more complex models.
  - **Robust shock resolution:** Roe's scheme and the Venkatakrishnan limiter ensure an accurate representation of shock waves and steep gradients.
- **Limitations:**
  - **Limited turbulence modelling:** The SA model may not fully resolve large-scale separated flows or secondary flow phenomena.
  - **High computational cost:** RANS simulations, especially on superfine grids, are significantly more expensive than Euler simulations.
  - **Convergence problems:** After reaching the fully developed flow structure, the residue starts to oscillate. The simulation does not reach residue values comparable to the Euler simulation. Looking at the distribution of residues in the domain, it has been hypothesised that this problem comes from the SBLI position, that keeps oscillating around the reflection point.

Grid convergence Mesh convergence was achieved after 4 refinements, with each run reaching a value of the residual on the density field below 10^{-7}. Due to computational limitations, it was not feasible to reach the same residual level as the Euler simulations. Therefore, while the results are robust and reliable, comparisons with the Euler solutions should be interpreted with caution and a critical perspective.
From Figure 9, it can be observed that the grid convergence is not as linear as in the Euler simulations. However, similar to the Euler case, all values of C_d lie within 1% of the finest solution. Notably, the finer grids after the first simulation deviate by less than 0.5%.

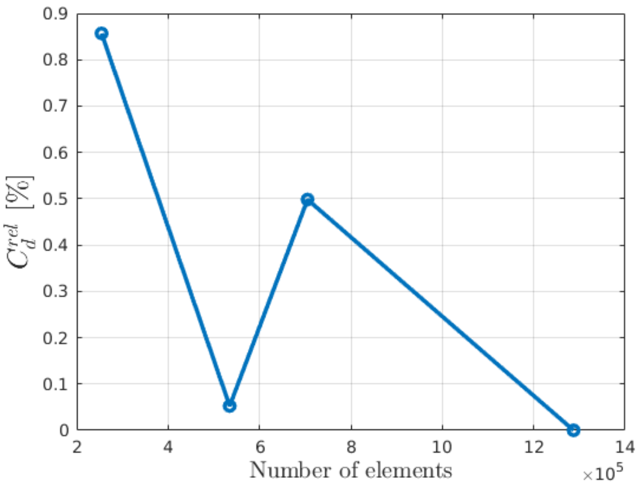
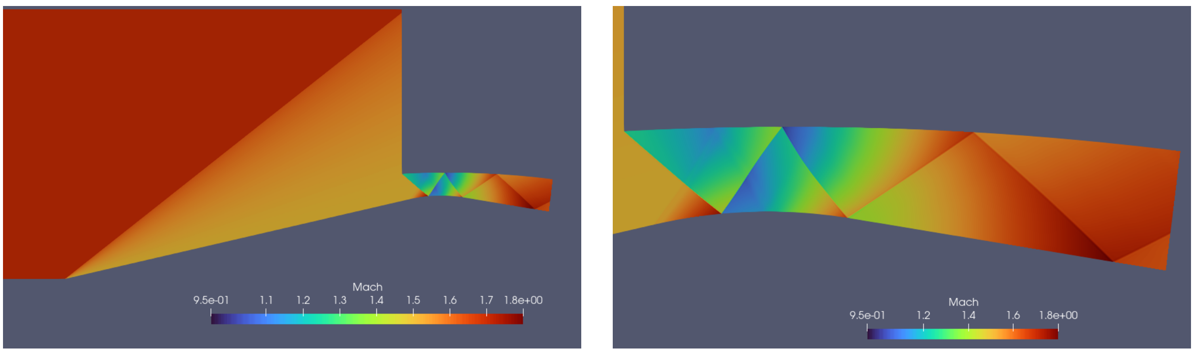


Figure 9: C_d percentage error relative to the finest mesh. RANS case.

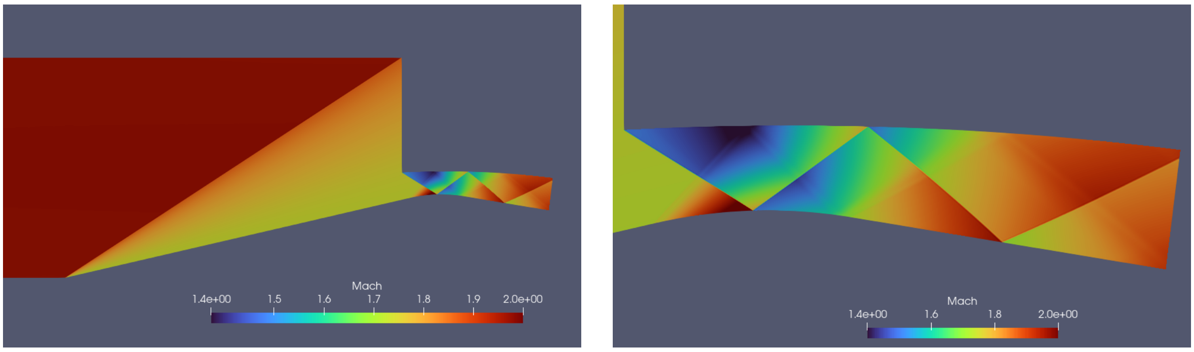
3. Results

3.1. Results of the Euler simulations

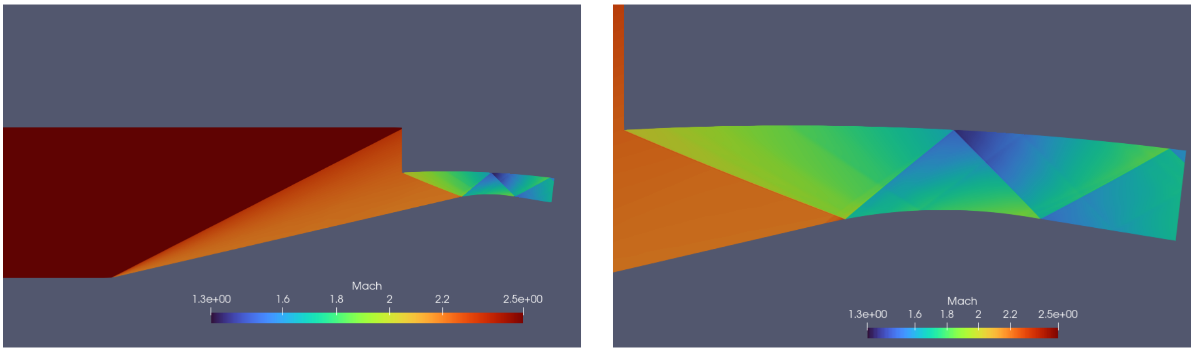
The shock patterns obtained via inviscid simulations are presented in Figure 10. The Mach number was chosen as the variable to highlight the flow structure.
It should be noted that each image includes its own colorbar. This choice was intentional, as it maximizes contrast and highlights the details of each inlet configuration, which might otherwise have been obscured.



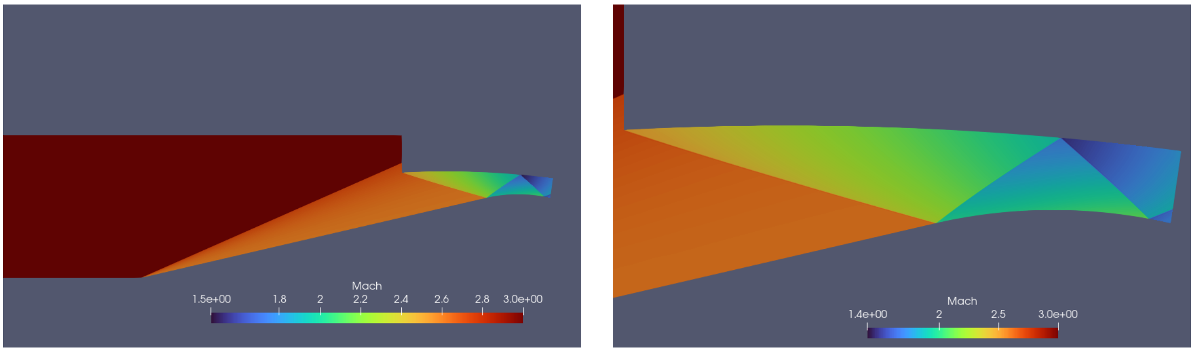
$M_\infty = 1.7$



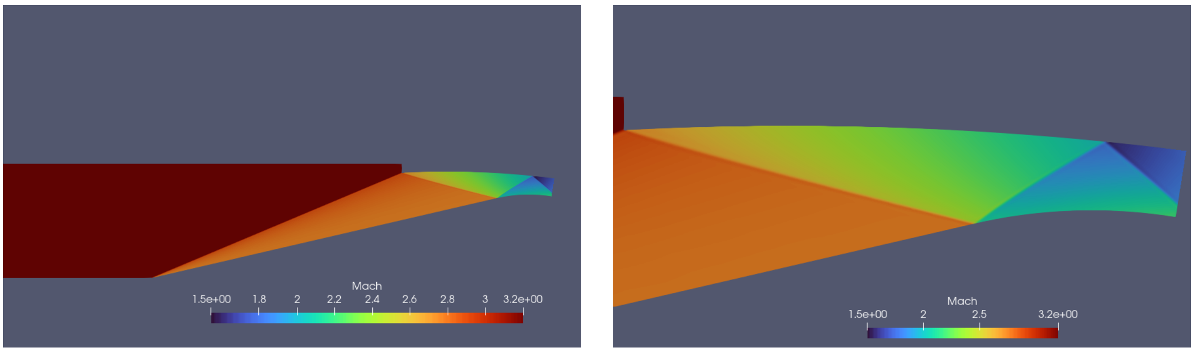
$M_\infty = 2.0$



$M_\infty = 2.5$



$M_\infty = 3.0$



$M_\infty = 3.2$

Figure 10: Results of the Euler simulations.

Looking at the results obtained in the last three configurations, it is clear that the flow greatly reduces its speed due to all the shock wave reflections. Focusing on the first two test cases, on the other hand, the flow counter-intuitively accelerates while still inside the duct, seemingly worsening the situation. It is important to remember, however, that immediately downstream with respect to the outlet should

occur the generation of a normal shock wave (due to the shock trap), whose position depends on the properties of the flow.
Upon closer inspection, it was found that the mean values of the flow, such as the Mach number, were extremely similar, in particular $M \approx 1.7$. This fact highlights the importance of the moving spike in guaranteeing the correct functioning of the shock trap bleed system, i.e. in maintaining the position of the normal shock wave constant and in providing consistent flow conditions to the engine at different flight regimes, limiting also the risk of unstart.
Furthermore, from these results it is possible to witness the complex interactions between shock waves and expansion fans, generated by the spike curvature. These phenomena are mostly visible at low Mach numbers due to high gradients and induced curvatures of shock lines. To further validate the inviscid solution, as visible in figure 11, the Mach number distribution along the spike and cowl surfaces is compared to that obtained by Coleville [4]. The results match closely.

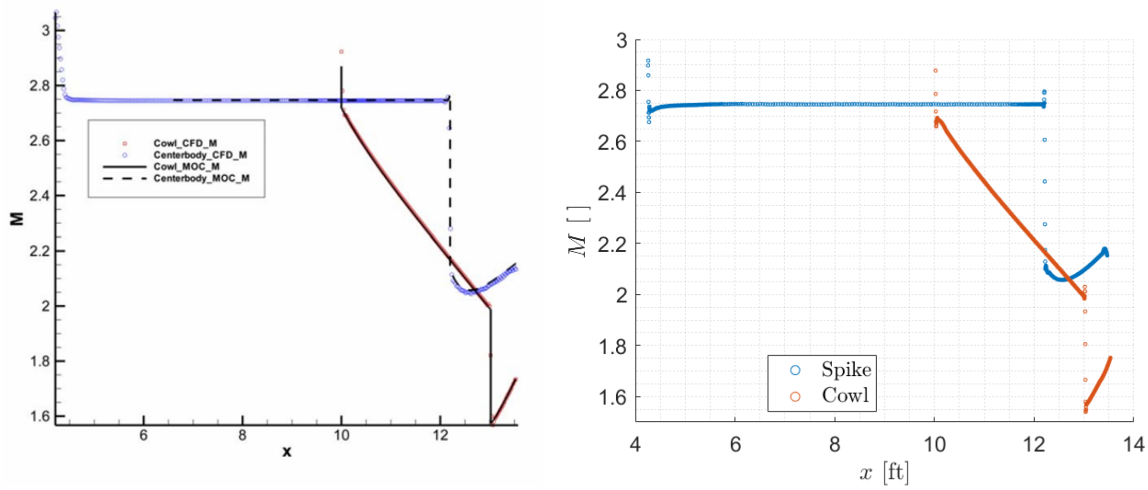


Figure 11: Comparison between Coleville's results (left) and those of the Euler simulation (right). Mach distribution along the spike and cowl surfaces at $M_\infty = 3.2$.

Another point of reference for validating the obtained results is Campbell's experimental work at NASA [7] on the inlet of the YF-12 aircraft, which can be considered the "predecessor" of the SR-71. The two aircraft share many features, including the design of the inlet. Extensive testing in a supersonic wind tunnel enabled them to determine the total pressure recovery of the inlet across various Mach numbers within the flight envelope, as shown in Figure 12. A similar graph was obtained from the Euler simulations investigating the Total Pressure Recovery (TPR) at the chosen values of Mach number, as represented in Figure 13.

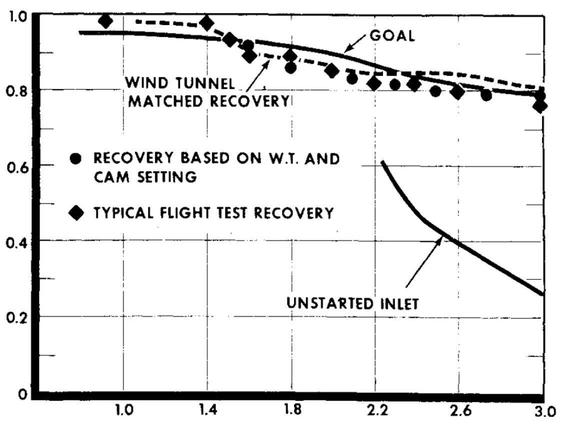


Figure 12: Experimental results of TPR [%] at different Mach numbers.
Courtesy of *NASA* [7].

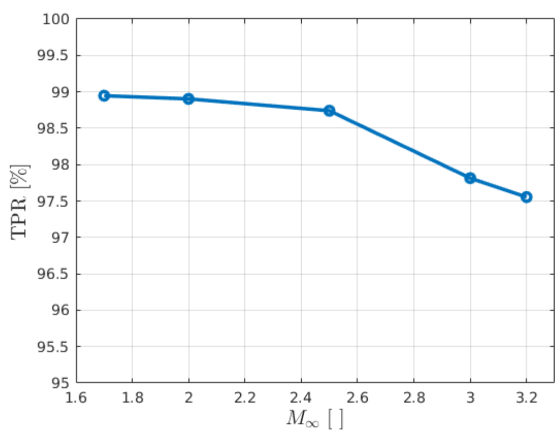


Figure 13: Euler TPR [%] results.

From the comparisons of the experimental and numerical results, there is a clear difference in absolute value of the percentage of recovery, but it can be seen that the trend in the Euler simulations closely resembles the one from the experiments, with a rather smooth decrease.
Furthermore, the experiments measured the outlet total pressure at the engine face, while the available data allowed modeling of the inlet only up to (but excluding) the shock trap. As a result, the contribution of the more intense and dissipative normal shock downstream was not accounted for.
Euler simulations accurately capture total pressure losses caused by discontinuities across shock waves, however the absence of viscosity may still pose a significant limitation for simulating other physical phenomena with the same level of accuracy.

3.2. Results of the RANS simulations

The results of the RANS simulation are presented in Figure 14, highlighting the key characteristics of viscous flows. These include the development of the boundary layer along the spike and cowl, as well as other viscous phenomena, such as Shock-Boundary Layer Interactions (SBLI) occurring at each shock reflection.
A small difference compared to the inviscid equations can be also found in the structure of the shocks, especially towards the outlet: the simulation resembles more the Euler case at Mach 3.0 instead of the one at 3.2, with a second shock reflection on the spike. This is probably due to the restriction of the actual inlet critical section, caused by the non-null boundary layer thickness.

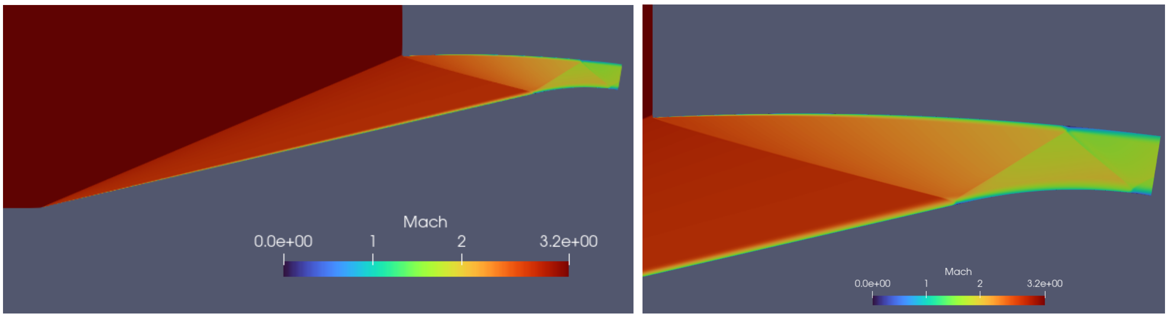


Figure 14: Result of RANS simulation at $M_\infty = 3.2$.

In Figure 15 it can be observed the focus on the first shock reflection on the spike. This allows to better appreciate the interaction between the boundary layer and the impinging shock wave, which leads to the formation of a small recirculation bubble, identifiable by the local increase in boundary layer thickness. Evidence of said bubble can be also found in Figure 16 through the representation of the streamlines, clearly rotating around a centre point.

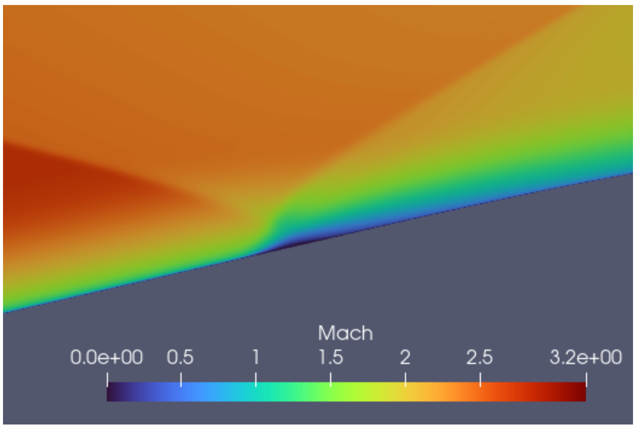


Figure 15: Detail of SBLI.

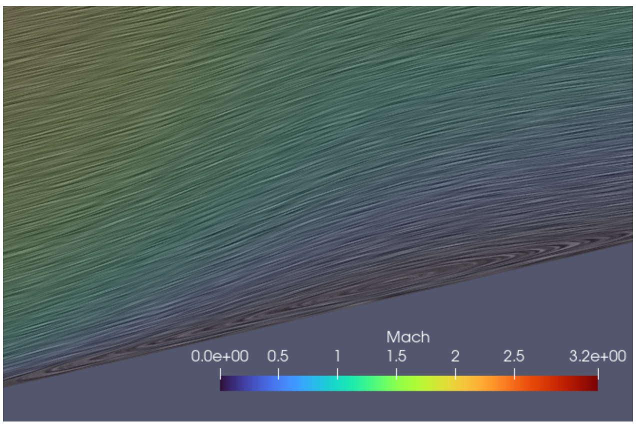


Figure 16: Zoom on recirculation bubble.

From a quantitative perspective, the presence of the boundary layer and viscous dissipation affects both

the mass flow rate and total pressure recovery: at Mach 3.2, the average mass flow rate at the outlet in the RANS simulation decreases slightly to 99.65% compared to the corresponding Euler case. In contrast, a notable discrepancy is observed in total pressure recovery, with the RANS result amounting to only 90.84% of the Euler case, as shown in Table 5.
This difference is imputable to the viscous effects of dissipation of energy in the boundary layer and at the points of SBLI.

Method	Outlet average P_{tot}	Total Pressure recovery
Euler	103697.5 Pa	97.56%
RANS	94277.5 Pa	88.62%
Experiments	[-]	$\sim$ 80%

Table 5: Results of the Total Pressure Recovery at Mach 3.2.

The total pressure recovery appear credible when compared to the experimental data from Campbell's study conducted in a supersonic wind tunnel [3], as summarized in Table 5.
While the RANS result may initially seem overestimated, it is important to remark that the experimental values are measured at the engine face, which lies outside the simulated domain. This implies the presence of additional sources of pressure loss not accounted for in the current computational analysis. If those factors were taken into consideration, it is highly likely that the Total Pressure Recovery in the RANS simulations would better approximate the experimental results.

3.3. Conclusion

The results of this study point to some interesting conclusions:

- The Euler equations are a valid way to describe internal flows for aeronautical applications, saving computational resources when compared with viscous simulations. Shock structures are predicted with sufficient accuracy and the mass flow error is negligible, making these simulations potentially useful to spot unstart conditions. However, they overestimate the total pressure recovery approximately by 10% compared to RANS, not taking into account viscous dissipation.
- Unlike Coleville's research [4], the RANS simulation does not actually reach an unstart condition. However, the turbulent boundary layer's (TBL) chaotic nature may represent a dangerous instability factor in case of unsteady perturbations during flight. Active boundary layer control systems, such as the spike suction bleed, can prevent this dangerous instance by reducing the TBL width and preventing separation.
- RANS simulations are capable of resolving complex flow structures like Shock-Boundary Layer Interactions, but these results are more qualitative and should be further studied and investigated.
- RANS simulations demonstrated a better representation of Total Pressure Recovery compared to Euler simulations and experimental data. However, it is important to emphasize that, due to inherent modelling limitations, the RANS equations may not fully capture the physical phenomena. As such, their results should not be quantitatively accepted without a thorough and critical evaluation.

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